

where, $x_1, x_2, \dots, x_n$ are n variables; $a_1, a_2, \dots, a_n$, and b are constants/parameters and at least one of $a_1, a_2, \dots, a_n$ is non-zero.

This equation represents a straight **line** in n -dimensional space.

Until and unless mentioned otherwise, all our variables will be real-valued ones, and constants and parameters also assume values which are only real numbers.

8.2.2 Solution of a Linear Equation

$$a_1 x_1 + a_2 x_2 + \dots + a_n x_n = b \quad \dots \dots \dots (I)$$

is a set of n constant real/ complex values c_j for $j = 1, 2, \dots, n$ such that when the variable x_j is replaced by the corresponding constant c_j in equation (I), then value of LHS becomes the constant b , which is the RHS of (I).

8.2.3 System of Linear Equations

If we are given the following two equations,

$$3x + 4y = 4 \quad \dots \dots \dots (II)$$

and

$$6x + y = 8 \quad \dots \dots \dots (III)$$

not as a system,

then the x (or, y) in (II) has **no relation with** the x (or, y) of (III).

And, then

$x=0$, and $y=1$ could have been a solution of (II),

and, at the same time

$x=1$, and $y=2$ could have been a solution of (III).

We could have very well written (II) and (III) equivalently as

$$3x + 4y = 4 \quad \dots \dots \dots (II)$$

and

$$6u + v = 8 \quad \dots \dots \dots (III')$$

However, when we say the **system of equations**

$$3x + 4y = 4 \quad \dots \dots \dots (II)$$

and

$$6x + y = 8 \quad \dots \dots \dots (III)$$

then we *intend the same value of x (and of y)* is to be substituted as solution in **both** the equations (II) and (III). In this case, (II) and (III) as a system, has unique solution obtained as follows

$$7y = 0 \quad (\text{obtained through 2 times of (II) - (III)})$$

implying

$y=0$ and $x = 4/3$.

In general, by a **system of m linear equations** in n variables $x_1, x_2, \dots, x_n$, we mean the following

$$\left. \begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1j}x_j + \dots + a_{1n}x_n &= b_1 \\ a_{i1}x_1 + a_{i2}x_2 + \dots + a_{ij}x_j + \dots + a_{in}x_n &= b_i \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mj}x_j + \dots + a_{mn}x_n &= b_m, \end{aligned} \right\} \quad (A)$$

for $i = 1, 2, \dots, m$, and $j = 1, 2, \dots, n$,

for which, we are searching for constants c_j 's such that

for the same value c_j of x_j with $j = 1, 2, \dots, n$,

all the equations in (A) are satisfied *simultaneously*.

This is why, sometime, we use the phrase ‘*System of Simultaneous Equations*’.

8.2.4 System of Homogeneous Linear Equations

The system of equations represented by (A) above is said to be **homogenous if**

$$b_i = 0, \text{ for all } i = 1, 2, \dots, m.$$

More explicitly, a homogeneous system of equations is of the form

$$\left. \begin{aligned} &a_{11}x_1 + a_{12}x_2 + \dots + a_{1j}x_j + \dots + a_{1n}x_n = 0 \\ &\quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \\ &a_{i1}x_1 + a_{i2}x_2 + \dots + a_{ij}x_j + \dots + a_{in}x_n = 0 \\ &\quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \\ &a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mj}x_j + \dots + a_{mn}x_n = 0, \end{aligned} \right\} \quad (\text{B})$$

for $i = 1, 2, \dots, m$, and $j = 1, 2, \dots, n$.

In the case of Homogeneous system of linear equations,

$$x_j = 0, \text{ for } j = 1, 2, \dots, n,$$

is *always* a solution.

Such a solution of homogeneous system of equations is called the *trivial solution*.

However, a homogeneous system of linear equations may also have other solutions, i. e. a solution in which at least one x_j has a non-zero value. Such a solution is called a ***non-trivial solution*** of the homogeneous system of linear equations.

8.3 REPRESENTATION OF SYSTEM OF EQUATIONS AS MATRIX EQUATION

For large values of natural numbers m and n , the system of equations under (A) of subsection 8.2.3 become quite unwieldy for comprehension and to carry out further operations for getting required solution.

One of the approaches to facilitate compact representation of a system of linear equations is through its **representation as a Matrix equation**.

Apart from providing compactification of representation of the system (A), the approach has the additional advantage of allowing use of theories and tools, developed for matrices in previous units, for solving system of equations (A).

(A) Matrix Representation of System of Equations

The system of linear equations (A) can be equivalently represented by the matrix

$$\mathbf{A} \times \mathbf{X} = \mathbf{B},$$

where $\mathbf{A}$, $\mathbf{X}$ and $\mathbf{B}$ are respectively $m \times n$ matrix, $n \times 1$ matrix and $m \times 1$ matrix; and further $\mathbf{A} \times \mathbf{X}$ is the matrix obtained by matrix multiplication of the matrices $\mathbf{A}$ and $\mathbf{X}$, as given below

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{1n} \\ \dots & \dots & \dots \\ a_{i1} & a_{ij} & a_{in} \\ \dots & \dots & \dots \\ a_{m1} & a_{mj} & a_{mn} \end{bmatrix}, \mathbf{X} = \begin{bmatrix} x_1 \\ \dots \\ x_j \\ \dots \\ x_n \end{bmatrix}, \mathbf{B} = \begin{bmatrix} b_1 \\ \dots \\ b_i \\ \dots \\ b_m \end{bmatrix}$$

(B) For solution, simplification of system of equations and corresponding matrix operations

The involved ideas are explained through the following example.

Example: Let us consider the following system of three linear equations in three variables

$$\left. \begin{array}{l} 3x + y + 2z = 11 \dots (\text{Eq.1}) \\ x + 2y - 2z = -1 \dots (\text{Eq.2}) \\ 2x - 2y - z = -5 \dots (\text{Eq.3}) \end{array} \right\} \dots (\text{C})$$

By the usual method of elimination of variables, or otherwise, the solution of the above system is given by: $x=1$, $y=2$, and $z=3$

Matrix representation of the above system is

$$\begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & -2 \\ 2 & -2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 11 \\ -1 \\ -5 \end{bmatrix} \dots (\text{D})$$

where matrices $\mathbf{A}$, $\mathbf{X}$ and $\mathbf{B}$ are given respectively by

$$A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & -2 \\ 2 & -2 & -1 \end{bmatrix} \quad x = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad B = \begin{bmatrix} 11 \\ -1 \\ -5 \end{bmatrix}$$

Even more concise representation of (C) is given by the following **augmented matrix representation**

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 11 \\ 1 & 2 & -2 & -1 \\ 2 & -2 & -1 & -5 \end{array} \right] \quad \dots (E)$$

where the column vector

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

is understood.

Check Your Progress 1

1) Tell, which of the following is a linear equation.

(i) $3x + 4y = 0$ (ii) $2x - 11$ (iii) $xy = 3$, (iv) $7 + 4 = 11$ (v) $x \cdot (x - 1) = 0$

.....

2) Tell which of the following is a homogeneous linear equation

(i) $x(x + 2) = 0$ (ii) $x - 4 + y = 0$ (iii) $x + 3y + 4z = 0$

.....

8.4 ELEMENTARY ROW OPERATIONS AND UNCHANGED SOLUTION SET FOR CORRESPONDING SYSTEM OF EQUATIONS

Now, it can be easily observed that the solution set remains unchanged if we interchange the order of any two equations in (C).

The corresponding matrix for changed order of equations, is obtained from (D) by interchanging corresponding rows in A and B simultaneously or, from (E) by interchanging corresponding rows.

For example, if we interchange (Eq. 1) and (Eq.3) in (C), we get

$$\left[\begin{array}{l} 2x - 2y - z = -5 \dots (\text{Eq.3}) \\ x + 2y - 2z = -1 \dots (\text{Eq.2}) \\ 3x + y + 2z = 11 \dots (\text{Eq.1}) \end{array} \right] \quad \dots (C')$$

Then corresponding Matrix representation is obtained by interchanging $\text{Row}_1 \leftrightarrow \text{Row}_3$ as shown below

$$\begin{bmatrix} 2 & -2 & -1 \\ 1 & 2 & -2 \\ 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -5 \\ -1 \\ 11 \end{bmatrix} \quad \dots (D')$$

And the corresponding augmented matrix representation is also obtained by interchanging $\text{Row}_1 \leftrightarrow \text{Row}_3$ as shown below

$$\left[\begin{array}{ccc|c} 2 & -2 & -1 & 5 \\ 1 & 2 & -2 & -1 \\ 3 & 1 & 2 & 11 \end{array} \right] \quad \dots (E')$$

From the above argument, we draw the following conclusions:

- interchange of rows in (E), or simultaneous interchange of the same rows in A and B of (D), do not change the solution set of corresponding system of equations.
- If we multiply all the involved constants of some of the equation in C by a non-zero constant, say k, throughout, then the equation, (and system of equations) remains the same as far as the solution is concerned.

The changed equation system is represented by the augmented matrix which is obtained from (E) by multiplying the corresponding row by k.

For example, by 2 x (Eq. 3) in (C), we get the system

$$\left. \begin{array}{l} 3x + y + 2z = 11 \dots (\text{Eq.1}) \\ x + 2y - 2z = -1 \dots (\text{Eq.2}) \\ 4x - 4y - 2z = -10 \dots (\text{Eq.3}') \end{array} \right\} \quad \dots (C'')$$

Such that the solution set of (C) and (C'') remains same.

Corresponding augmented matrix is

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 11 \\ 1 & 2 & -2 & -1 \\ 4 & -4 & -2 & 10 \end{array} \right] \quad \dots (E'')$$

We may now say that if we multiply the elements of a row by a non-zero constant k in (i) augmented matrix (E), or in (ii) the corresponding row of matrices A and B simultaneously, then solution set remains unchanged.

Further, if we add corresponding terms of one equation, say, Eq.' of (C) to the corresponding terms of another equation, say Eq.'' of (C), then the new system of equations has the same solution set as the earlier one. The changed equation system is represented by the augmented matrix which is obtained from (E) by adding row corresponding to Eq.' to the row corresponding to Eq.'.

Combined with the conclusion (b), we may further say that, if we add to a row the k(non-zero) multiples of another row in (i) the corresponding augmented matrix (E), or (ii) the corresponding rows of matrices A and B simultaneously, then solution set remains unchanged.

8.4.1 Elementary Row Operations and Unchanged Solution Set

The following type of operations on the rows of a given matrix, say M , are called **Elementary Row operations**

- i) Interchange of Rows of matrix M ; $R_i \leftrightarrow R_j$,
- ii) Multiplying elements of a row by some non-zero constant k ; $R_i \rightarrow k R_i$
- iii) Adding k (non-zero) times multiples of elements of a row to the corresponding elements of another row of M . $R_i \rightarrow R_i + k R_j$

Theorem A: Elementary row operations on the Augmented matrix (or the matrices A and B of the matrix equation $AX = B$) do not change the solution set of the system of equations represented by the matrix/matrices.

Advantage/ Application of Theorem A: Without going through the cumbersome process of scanning the systems of equations repeatedly, which are obtained, at various stages, we just examine the corresponding Augmented matrix (or, the corresponding matrix equation) for simplifying the system through row operations, for getting the final solution(s), if exists (exist). This is explained through solving the

$$\left. \begin{array}{l} 3x + y + 2z = 11 \dots (\text{Eq.1}) \\ x + 2y - 2z = -1 \dots (\text{Eq.2}) \\ 2x - 2y - z = -5 \dots (\text{Eq.3}) \end{array} \right\} \dots (\text{C})$$

and using the corresponding augmented matrix

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 11 \\ 1 & 2 & -2 & -1 \\ 2 & -2 & -1 & -5 \end{array} \right] \dots (\text{E})$$

Simulating—the usual method of elimination of variables by making zero the corresponding coefficients—through row operations, as follows

$$R_1 \rightarrow R_1 - 3 R_2$$

and

$$R_3 \rightarrow R_3 - 2 R_2, \text{ we get}$$

$$\left[\begin{array}{ccc|c} 0 & -5 & 8 & 14 \\ 1 & 2 & -2 & -1 \\ 0 & -6 & 3 & -3 \end{array} \right] \dots (\text{E}^{\text{iii}})$$

$$\text{By } R_1 \leftrightarrow R_2 \text{ we get}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -2 & 1 \\ 0 & -5 & 8 & 14 \\ 0 & -6 & 3 & -3 \end{array} \right] \dots (\text{E}^{\text{iv}})$$

$$\text{By } R_3 \rightarrow 5 R_3$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -2 & 1 \\ 0 & -5 & 8 & 14 \\ 0 & -30 & 15 & -15 \end{array} \right] \dots (\text{E}^{\text{v}})$$

By $R_3 \rightarrow R_3 - 6 R_2$, we get

$$\left[\begin{array}{ccc|c} 1 & 2 & -2 & -1 \\ 0 & -5 & 8 & 14 \\ 0 & 0 & -33 & 99 \end{array} \right] \quad \dots (E^{vi})$$

By $R_3 \rightarrow -33 R_3$, we get

$$\left[\begin{array}{ccc|c} 1 & 2 & -2 & -1 \\ 0 & -5 & 8 & 14 \\ 0 & 0 & 1 & 3 \end{array} \right] \quad \dots (E^{vii})$$

For getting a solution, this is the usual process in which elements below the main diagonal of augmented matrix are attempted to be made zeros.

From R_3 , we get

$$z = 3,$$

using it in R_2 , we get

$$-5y + 8(3) = 14 \text{ implying}$$

$$y = 2$$

Using $z=3$ and $y=2$ in R_1

we get

$$1.x + 2(2) - 2(3) = -1$$

implying $x=1$

Thus, we get $x=1$, $y=2$, and $z=3$ as solution of the given system of equations.

8.5 LINEAR INDEPENDENCE AND SYSTEM OF EQUATIONS

Next, we consider the case, when some equation in the given set of equations, can be obtained as linear combination of some of the other.

8.5.1 Linear Combination and Linear Dependence of Equations

Continuing with our system of equations

$$\left. \begin{array}{l} 3x + y + 2z = 11 \dots (\text{Eq.1}) \\ x + 2y - 2z = -1 \dots (\text{Eq.2}) \\ 2x - 2y - z = -5 \dots (\text{Eq.3}) \end{array} \right\} \quad \dots (C)$$

with the corresponding augmented matrix

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 11 \\ 1 & 2 & -2 & -1 \\ 2 & -2 & -1 & -5 \end{array} \right] \quad \dots (E)$$

Next, by replacing (Eq. 3) in the system (C), by *the equation* which corresponds to $R_3 \rightarrow 2 R_1 + 3 R_2$

$$\text{is } 2(3x + y + 2z) + 3(x + 2y - 2z) = 2(11) + 3(-1),$$

$$\text{i.e., } 9x + 8y - 2z = 19.$$

Thus, the new system of equations becomes

$$\left[\begin{array}{l} 3x + y + 2z = 11 \dots (\text{Eq.1}) \\ x + 2y - 2z = -1 \dots (\text{Eq.2}) \\ 9x + 8y - 2z = 19 \dots (\text{Eq.4}) \end{array} \right] \quad \dots (F)$$

And corresponding augmented matrix is

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 11 \\ 1 & 2 & -2 & -1 \\ 9 & 8 & -2 & -19 \end{array} \right] \quad \dots (G)$$

As discussed earlier, solution set does not change by elementary row operations on augmented matrix, the solution set of (F) also corresponds to (G') obtained from (G) by

$R_3 \rightarrow R_3 - (2R_1 + 3R_2)$, we get

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 11 \\ 1 & 2 & -2 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \dots (G')$$

But, the last row of the augmented matrix corresponds to the equation

$$0x + 0y + 0z = 0,$$

which does not yield any (new) information. Thus, augmented matrices (G) and (G') correspond actually to not three equations, but at most two equations only, viz.

$$3x + y + 2z = 11 \quad \dots (\text{Eq.1})$$

$$x + 2y - 2z = -1 \quad \dots (\text{Eq.2})$$

The equation (Eq. 4) is actually

$$(\text{Eq. 4}) = 2 (\text{Eq. 1}) + 3 (\text{Eq. 2}) \quad \dots \dots (P)$$

And does not add to available information for solution.

In cases like (P) above, when, in actuality, we have 2 equations in 3 variables, the number of solutions is infinite as shown below.

By (Eq.1) – 3 (Eq. 2), we get

$$-5y + 8z = 14$$

Or

$$y = (8z - 14) / 5 \quad \dots \dots (Q)$$

Similarly, by (Eq.2) – 2 (Eq. 1), we get

$$-5x - 6z = -23.$$

Or

$$x = (23 - 6z) / 5 \quad \dots \dots (R)$$

By giving values 1, 2, 3, ... to z,

we get corresponding infinitely set of values for x and y.

Thus, the system of three equations (Eq. 1), (Eq. 2) and (Eq. 4), has only two linearly independent equations, which may be chosen as (Eq. 1), and (Eq. 2). ***And these have infinitely many solutions.*** In general,

The relation (P) above represents the case of an equation, e. g. Eq. 4, being linear combination of the two equations, (Eq. 1) and (Eq. 2).

In general, if an equation (Eq. k) can be represented as

$$(\text{Eq. } k) = \lambda_1 (\text{Eq. } 1) + \lambda_2 (\text{Eq. } 2) + \dots + \lambda_{(k-1)}(\text{Eq. } (k-1)) + \lambda_{(k+1)}(\text{Eq. } (k+1)) + \dots + \lambda_n (\text{Eq. } n),$$

for constants λ_j , then

- i) (Eq. k) is said to be *linear combination* of equations (Eq. 1), (Eq. 2), Eq. (k-1)), (Eq. (k+1)), ... (Eq. n), and
- ii) (Eq. k)—if added to a system of equations containing (Eq. 1), (Eq. 2), Eq. (k-1)), (Eq. (k+1)), ... (Eq. n)—*does not contribute any additional information* toward unique solution of the system of equations.

Linear combination of equations leads to the concepts of linear dependence.

8.5.2 Linear Independence of m Equations in n Variables

(A) Definitions

- i) A set of m equations, say, (Eq. 1), (Eq. 2), ... & (Eq. m) in n variables, where $m \leq n$, is said to be *linearly dependent* if, for some j with for $1 \leq j \leq m$, the equation, say (Eq. j) can be written as a linear combination of the other equations. For example, the following set (F) of equations

$$\left. \begin{array}{l} 3x + y + 2z = 11 \dots (\text{Eq.1}) \\ x + 2y - 2z = -1 \dots (\text{Eq.2}) \\ 9x + 8y - 2z = 19 \dots (\text{Eq.4}) \end{array} \right\} \dots (\text{F})$$

is a set of linearly dependent equations, because (Eq. 4) = 2 (Eq. 1) + 3 (Eq. 2).

- ii) A set of m equations, say, (Eq. 1), (Eq. 2), ... & (Eq. m) in n variables, where $m \leq n$, is said to be *linearly independent* if, for no j with $1 \leq j \leq m$, the equation, say (Eq. j) can be written as a linear combination of the other equations.

For example, our original set of equations

$$\left. \begin{array}{l} 3x + y + 2z = 11 \dots (\text{Eq.1}) \\ x + 2y - 2z = -1 \dots (\text{Eq.2}) \\ 2x + 2y - z = -5 \dots (\text{Eq.3}) \end{array} \right\} \dots (\text{C})$$

is a set of 3 linearly independent equations, in the sense that none of the three equations can be written as a linear combination of the other two.

For establishing the last claim, we have the following test, which we state without proof.

(B) Test for Linear independence of m equations in n variables, with $m \leq n$.

In the augmented matrix representation of the system of equations, while attempting to make the elements below the main diagonal zeros, if we get one row of all-zeros, then the equations are Linearly Dependent, otherwise the given set of equations is linearly independent.

While solving the system of equations (C), through various elementary row operations, we obtained

$$\left[\begin{array}{ccc|c} 1 & 2 & -2 & -1 \\ 0 & -5 & 8 & 14 \\ 0 & 0 & -33 & -99 \end{array} \right] \quad \dots (E^{vi})$$

The last row containing zeros under main diagonal, is not all-zeros. Hence, the set of equations is linearly independent.

Now we state **some important results** for practical applications

Theorem B: If a set of n equations in n variables has some equation which is a linear combination of (or linearly dependent on) the remaining equations, **then the number of solutions is infinite.**

Generalizing further, we get

Theorem C: If we a system of m equations in n variables with $m < n$, then the number of solutions of the system is infinite.

8.6 CONSISTENCY OF A SYSTEM OF m LINEAR EQUATIONS IN n VARIABLES

In the previous section, we discussed the case of system of m equations in n variables, where $m \leq n$. The major question there was about linearly dependent/independent equations, i.e., about set of equations being superfluous or not.

Next, we discuss the reverse question: when $m > n$, i.e., m , the number of given equations is more than n , the number of variables.

Here the major concern/question is consistency, or inconsistency: whether the set of m linear equations can actually have a (common) solution or not.

Consider our earlier set (C) of equations:

$$\left. \begin{array}{l} 3x + y + 2z = 11 \dots (\text{Eq.1}) \\ x + 2y - 2z = -1 \dots (\text{Eq.2}) \\ 2x - 2y - z = -5 \dots (\text{Eq.3}) \end{array} \right] \quad \dots (C)$$

which we have already discussed has a solution $x = 1$, $y = 2$, and $z = 3$.

Next, we consider the enhanced system of linear equations, by adding the equation

$$4x - 3y + 2z = -2 \quad \dots \dots (\text{Eq. 5})$$